Causal Analysis Report

bpic12-a.xes.gz-flw

Decision points: 73936

Choice sets: 1

Transitions fitted: 10

Significant (p<0.05): 0

Generated: 2026-06-19 15:21

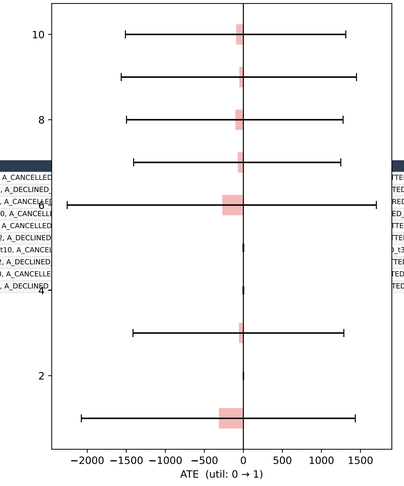
Step 1 — Association test (logistic regression + permutation)
H₀: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

(no data)

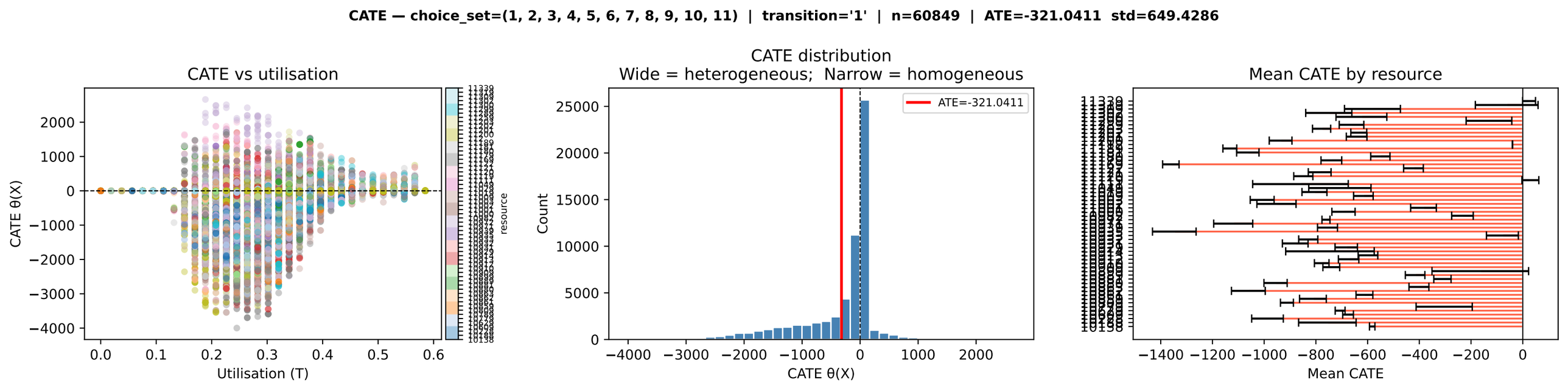
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

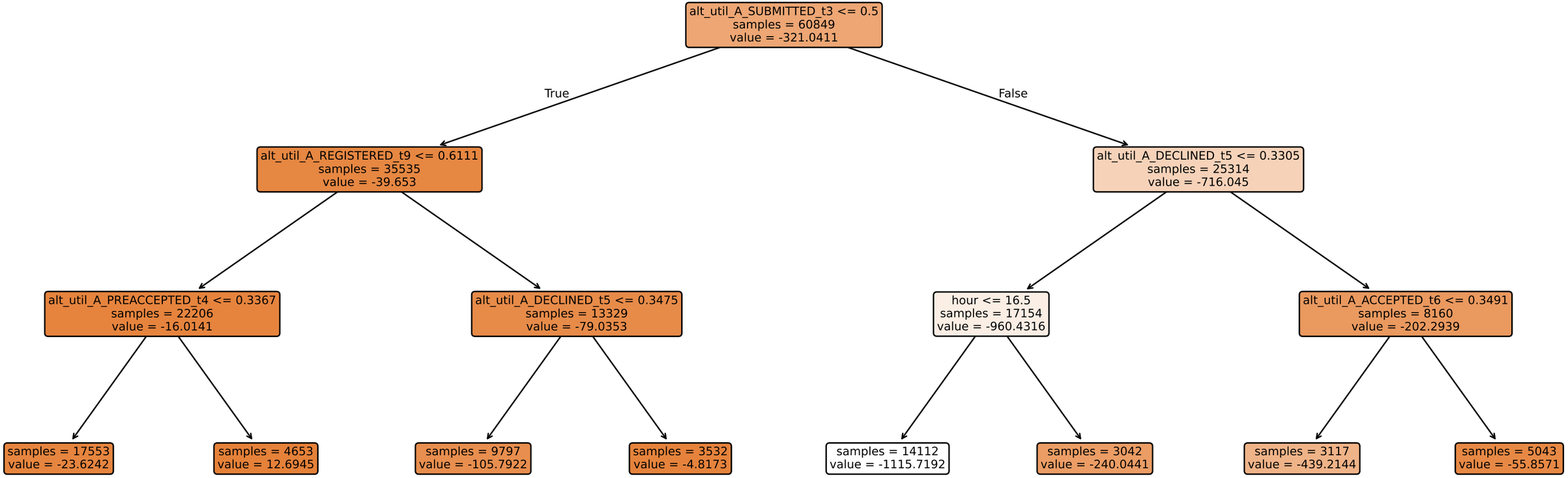
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	1	A_FINALIZED_t1	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED_t5, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_REGISTERED_t9, A_SUBMITTED_t3	-0.129842	0.655122	[-1.4139, 1.1542]	0.8429	False	θ=-0.1298: 1-unit ↑ util ↓ P(t1/A_FINALIZED_t1) by 13.0%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	2	A_CANCELLED_t2	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_DECLINED_t5, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_REGISTERED_t9, A_SUBMITTED_t3	0.201166	0.062611	[0.0784, 0.3239]	0.0013	True	θ=+0.2012: 1-unit ↑ util ↑ P(t2/A_CANCELLED_t2) by 20.1%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3	A_SUBMITTED_t3	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED_t5, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_REGISTERED_t9	0.282778	0.259874	[-0.2266, 0.7921]	0.2765	False	θ=+0.2828: 1-unit ↑ util ↑ P(t3/A_SUBMITTED_t3) by 28.3%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	4	A_PREACCEPTED_t4	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED_t5, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_REGISTERED_t9, A_SUBMITTED_t3	-0.24506	0.111547	[-0.4637, -0.0264]	0.028	True	θ=-0.2451: 1-unit ↑ util ↓ P(t4/A_PREACCEPTED_t4) by 24.5%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	5	A_DECLINED_t5	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_REGISTERED_t9, A_SUBMITTED_t3	-0.001933	0.128723	[-0.2542, 0.2504]	0.988	False	θ=-0.0019: 1-unit ↑ util ↓ P(t5/A_DECLINED_t5) by 0.2%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	6	A_ACCEPTED_t6	A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED_t5, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_REGISTERED_t9, A_SUBMITTED_t3	-0.224429	0.65914	[-1.5163, 1.0675]	0.7335	False	θ=-0.2244: 1-unit ↑ util ↓ P(t6/A_ACCEPTED_t6) by 22.4%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	7	A_PARTLYSUBMITTED_t7	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED_t5, A_FINALIZED_t1, A_PREACCEPTED_t4, A_REGISTERED_t9, A_SUBMITTED_t3	0.297091	0.260228	[-0.2130, 0.8071]	0.2536	False	θ=+0.2971: 1-unit ↑ util ↑ P(t7/A_PARTLYSUBMITTED_t7) by 29.7%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	8	A_ACTIVATED_t8	A_ACCEPTED_t6, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED_t5, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_REGISTERED_t9, A_SUBMITTED_t3	-0.341456	0.508849	[-9.1788, 8.4959]	0.9396	False	θ=-0.3415: 1-unit ↑ util ↓ P(t8/A_ACTIVATED_t8) by 34.1%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	9	A_REGISTERED_t9	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED_t5, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_SUBMITTED_t3	0.38328	4.501797	[-8.4402, 9.2068]	0.9322	False	θ=+0.3833: 1-unit ↑ util ↑ P(t9/A_REGISTERED_t9) by 38.3%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'τ_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	10	A_APPROVED_t10	A_ACCEPTED_t6, A_ACTIVATED_t8, A_CANCELLED_t2, A_DECLINED_t5, A_FINALIZED_t1, A_PARTLYSUBMITTED_t7, A_PREACCEPTED_t4, A_REGISTERED_t9, A_SUBMITTED_t3	0.327412	4.519939	[-8.5317, 9.1865]	0.9423	False	θ=+0.3274: 1-unit ↑ util ↑ P(t10/A_APPROVED_t10) by 32.7%

Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

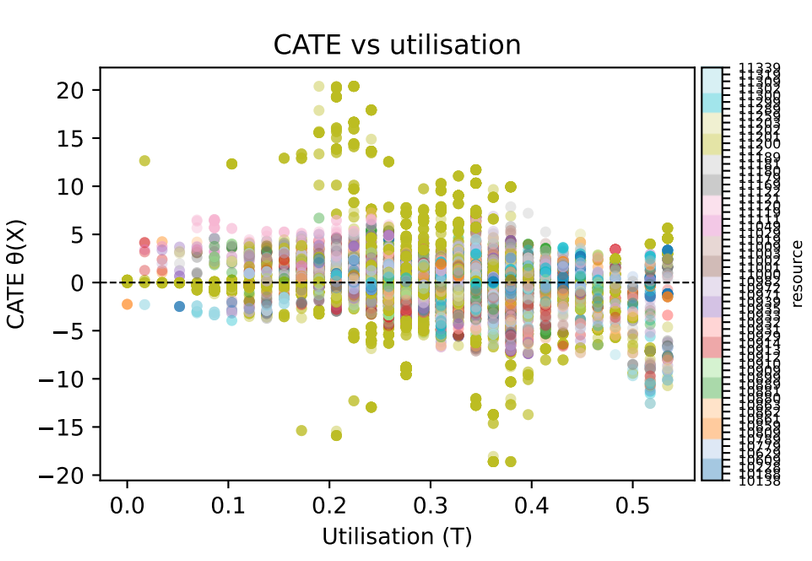


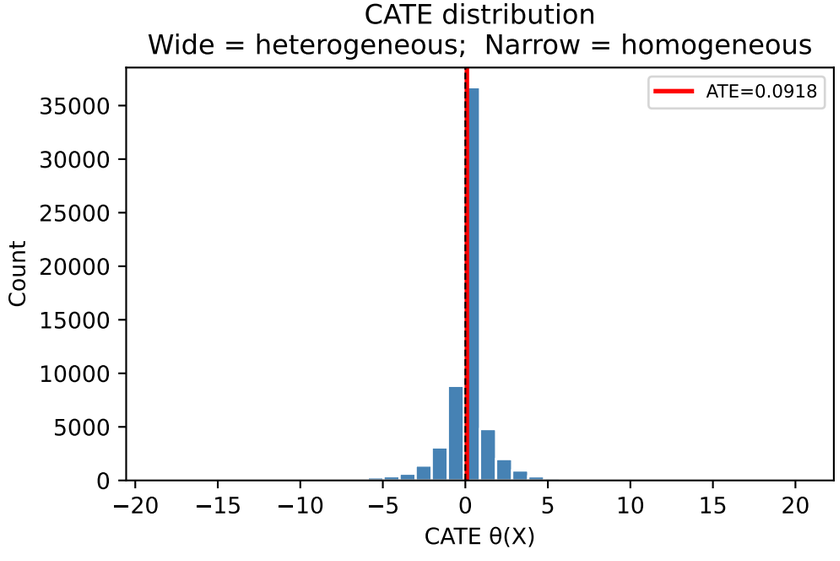
choice_set	choice_set_labels	n	choices_seen	transition	transition_label		ate	ci_95	p_value	reject_H0	note
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	1	A_FINALIZED_t1	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED	TED_t3	-321.041141	[-2076.0791, 1433.9968]	0.7199	False	ATE=-321.0411: 0→1 util ↓ P(t1/A_FINALIZED_t1) by 32104.1%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', 'A_DECLINED_	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	2	A_CANCELLED_t2	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_DECLINED_	TED_t3	0.091764	[-2.6550, 2.8385]	0.9478	False	ATE=+0.0918: 0→1 util ↑ P(t2/A_CANCELLED_t2) by 9.2%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3	A_SUBMITTED_t3	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLE	RED_t9	-63.40497	[-1414.1118, 1287.3018]	0.9267	False	ATE=-63.4050: 0→1 util ↓ P(t3/A_SUBMITTED_t3) by 6340.5%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	4	A_PREACCEPTED_t4	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLE	ED_t3	-0.086973	[-4.9459, 4.7720]	0.972	False	ATE=-0.0870: 0→1 util ↓ P(t4/A_PREACCEPTED_t4) by 8.7%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	5	A_DECLINED_t5	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLE	TED_t3	0.043774	[-4.7156, 4.8031]	0.9856	False	ATE=+0.0438: 0→1 util ↑ P(t5/A_DECLINED_t5) by 4.4%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	6	A_ACCEPTED_t6	A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED	TED_t3	-276.645357	[-2257.3745, 1704.0838]	0.7843	False	ATE=-276.6454: 0→1 util ↓ P(t6/A_ACCEPTED_t6) by 27664.5%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	7	A_PARTLYSUBMITTED_t7	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCE	_t3	-78.73539	[-1405.8653, 1248.3945]	0.9074	False	ATE=-78.7354: 0→1 util ↓ P(t7/A_PARTLYSUBMITTED_t7) by 7873.5%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	8	A_ACTIVATED_t8	A_ACCEPTED_t6, A_APPROVED_t10, A_CANCELLED_t2, A_DECLINED	TED_t3	-109.446553	[-1496.7256, 1277.8325]	0.8771	False	ATE=-109.4466: 0→1 util ↓ P(t8/A_ACTIVATED_t8) by 10944.7%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	9	A_REGISTERED_t9	A_ACCEPTED_t6, A_ACTIVATED_t8, A_APPROVED_t10, A_CANCELLE	TED_t3	-57.844761	[-1564.9290, 1449.2395]	0.94	False	ATE=-57.8448: 0→1 util ↓ P(t9/A_REGISTERED_t9) by 5784.5%
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	('A_FINALIZED_t1', 'A_CANCELLED_t2', 'A_SUBMITTED_t3', 'A_PREACCEPTED_t4', 'A_DECLINED_t5', 'A_ACCEPTED_t6', 'A_PARTLYSUBMITTED_t7', 'A_ACTIVATED_t8', 'A_REGISTERED_t9', 'A_APPROVED_t10', '<t_t11')	60849	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	10	A_APPROVED_t10	A_ACCEPTED_t6, A_ACTIVATED_t8, A_CANCELLED_t2, A_DECLINED_	TED_t3	-99.534407	[-1511.5194, 1312.4506]	0.8901	False	ATE=-99.5344: 0→1 util ↓ P(t10/A_APPROVED_t10) by 9953.4%

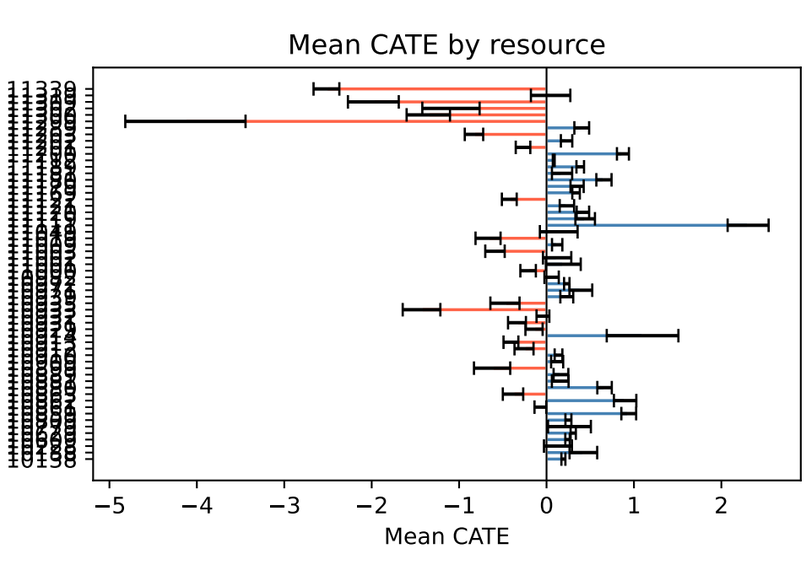

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='1' | n=60849 | ATE=-321.0411 std=649.4286
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-321.0411
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

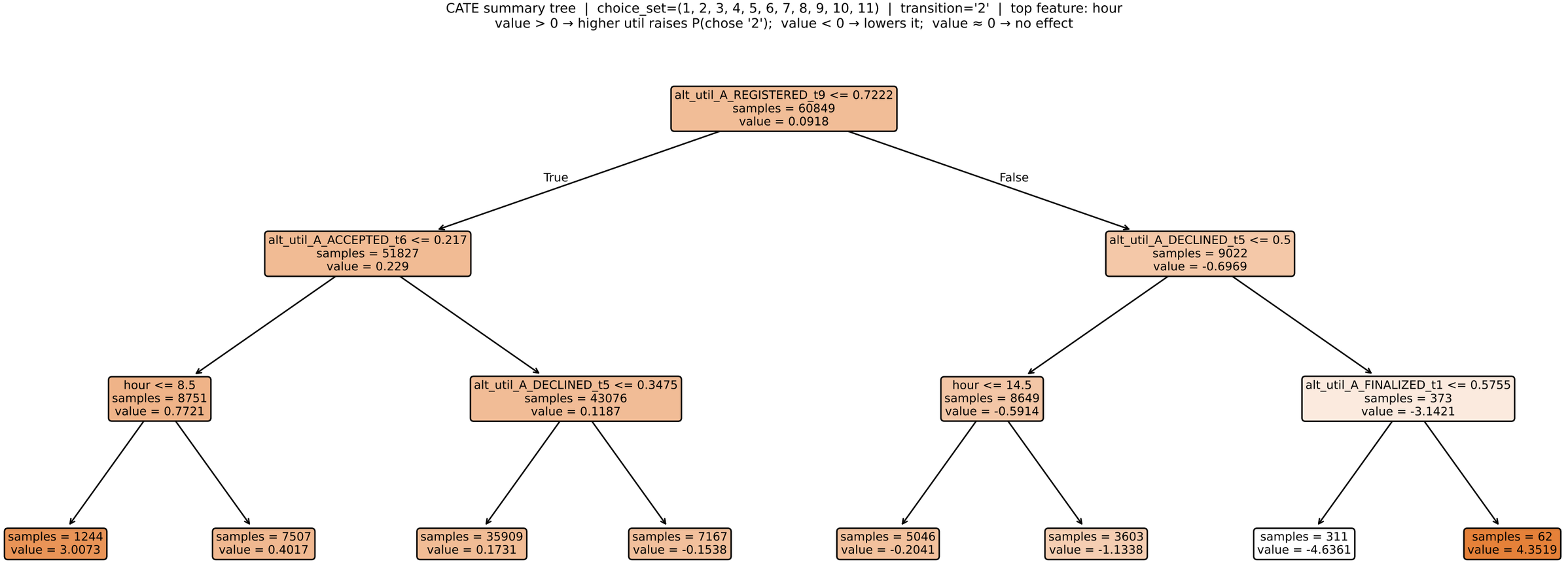

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='1' | top feature: alt_util_A_SUBMITTED_t3
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_SUBMITTED_t3 <= 0.5
samples = 60849
value = -321.0411
True
False
alt_util_A_REGISTERED_t9 <= 0.6111
samples = 35535
value = -39.653
alt_util_A_DECLINED_t5 <= 0.3305
samples = 25314
value = -716.045
alt_util_A_PREACCEPTED_t4 <= 0.3367
samples = 22206
value = -16.0141
alt_util_A_DECLINED_t5 <= 0.3475
samples = 13329
value = -79.0353
hour <= 16.5
samples = 17154
value = -960.4316
alt_util_A_ACCEPTED_t6 <= 0.3491
samples = 8160
value = -202.2939
samples = 17553
value = -23.6242
samples = 4653
value = 12.6945
samples = 9797
value = -105.7922
samples = 3532
value = -4.8173
samples = 14112
value = -1115.7192
samples = 3042
value = -240.0441
samples = 3117
value = -439.2144
samples = 5043
value = -55.8571

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='2' | n=60849 | ATE=0.0918 std=1.8909

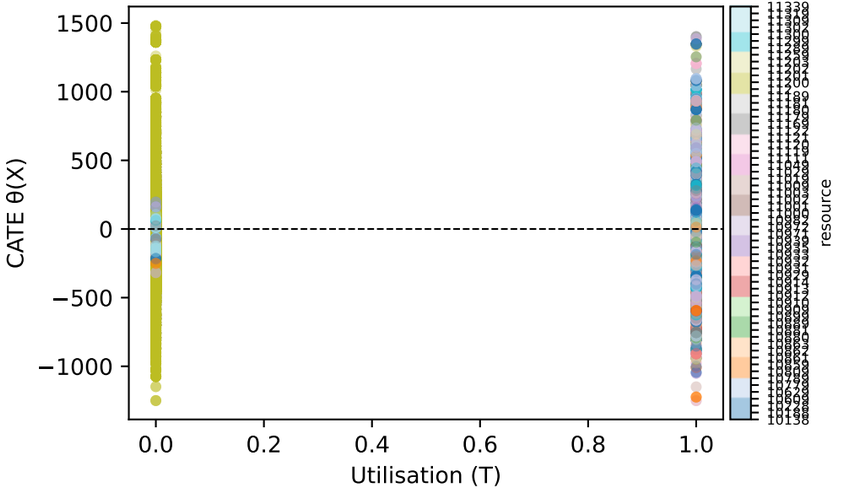


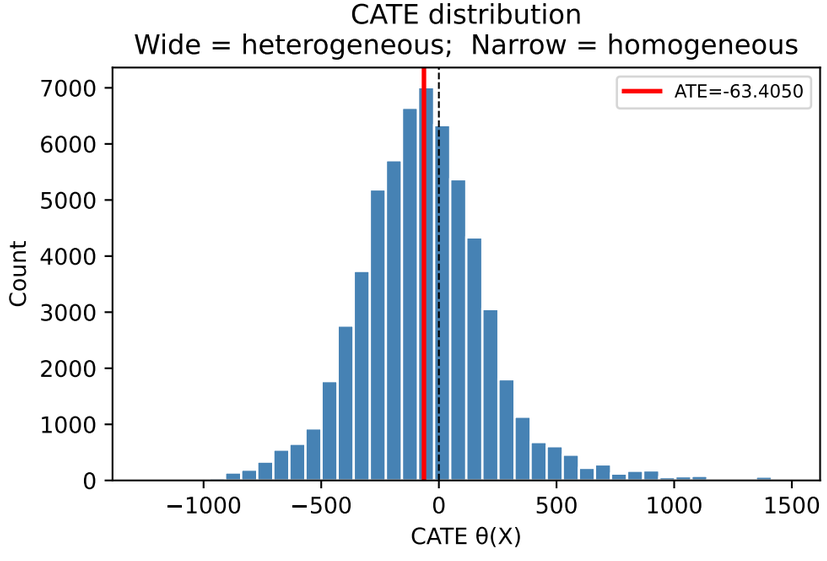


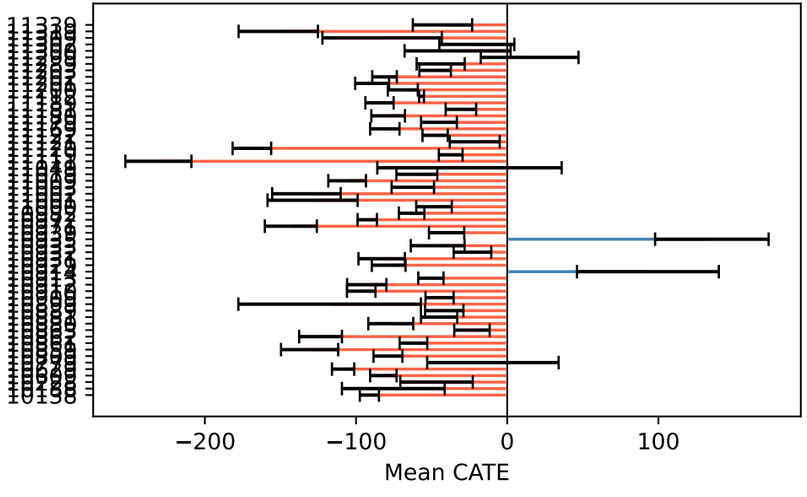


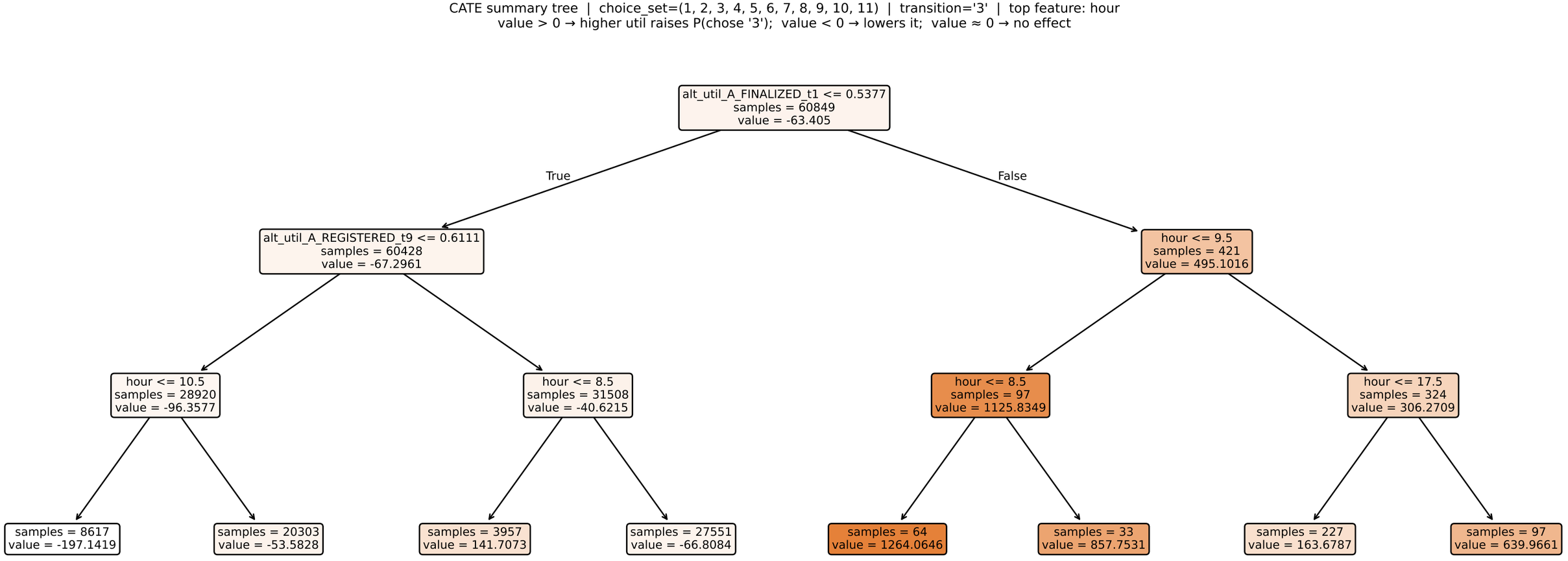

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='2' | top feature: hour
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_REGISTERED_t9 <= 0.7222
samples = 60849
value = 0.0918
True
False
alt_util_A_ACCEPTED_t6 <= 0.217
samples = 51827
value = 0.229
alt_util_A_DECLINED_t5 <= 0.5
samples = 9022
value = -0.6969
hour <= 8.5
samples = 8751
value = 0.7721
alt_util_A_DECLINED_t5 <= 0.3475
samples = 43076
value = 0.1187
hour <= 14.5
samples = 8649
value = -0.5914
alt_util_A_FINALIZED_t1 <= 0.5755
samples = 373
value = -3.1421
samples = 1244
value = 3.0073
samples = 7507
value = 0.4017
samples = 35909
value = 0.1731
samples = 7167
value = -0.1538
samples = 5046
value = -0.2041
samples = 3603
value = -1.1338
samples = 311
value = -4.6361
samples = 62
value = 4.3519

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='3' | n=60849 | ATE=-63.4050 std=285.2864

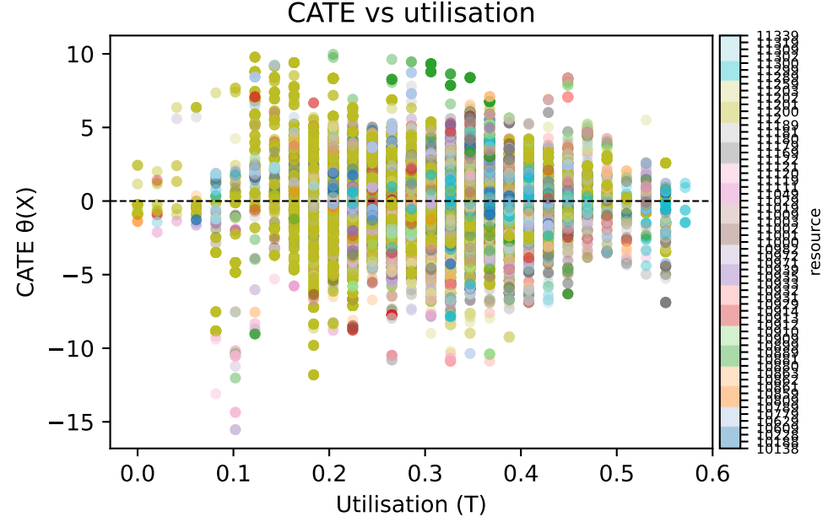


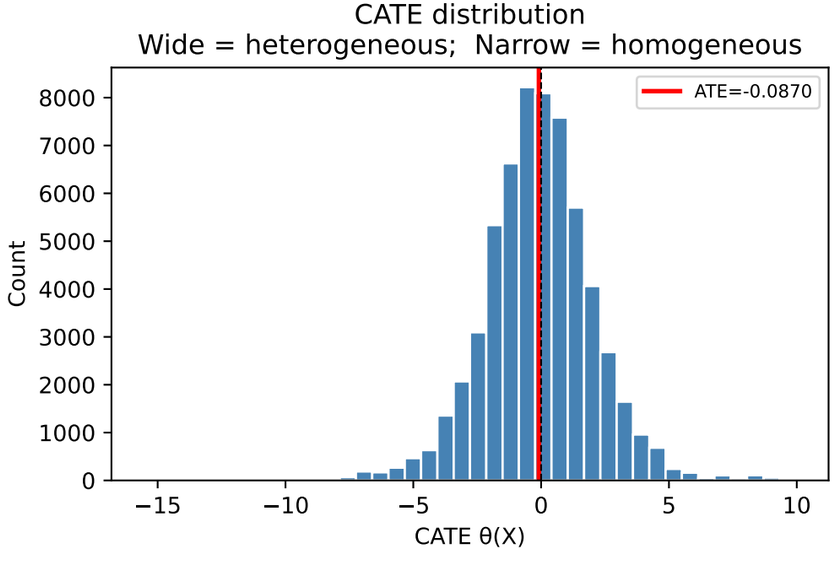


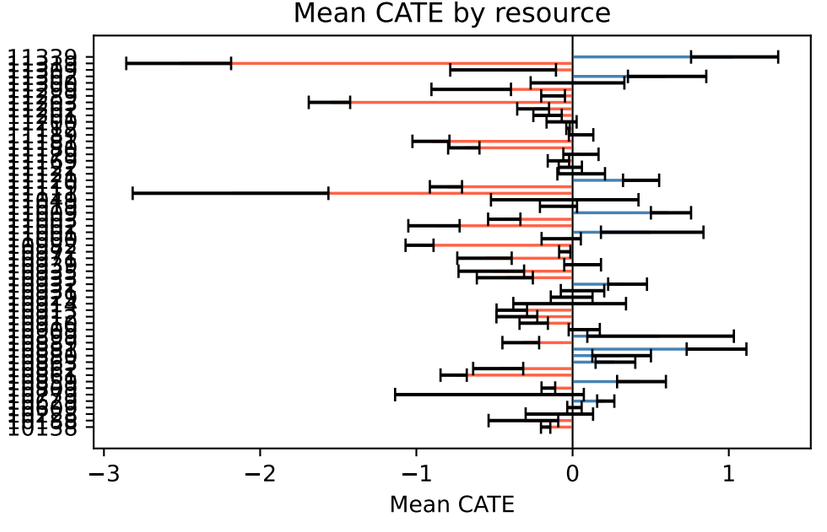


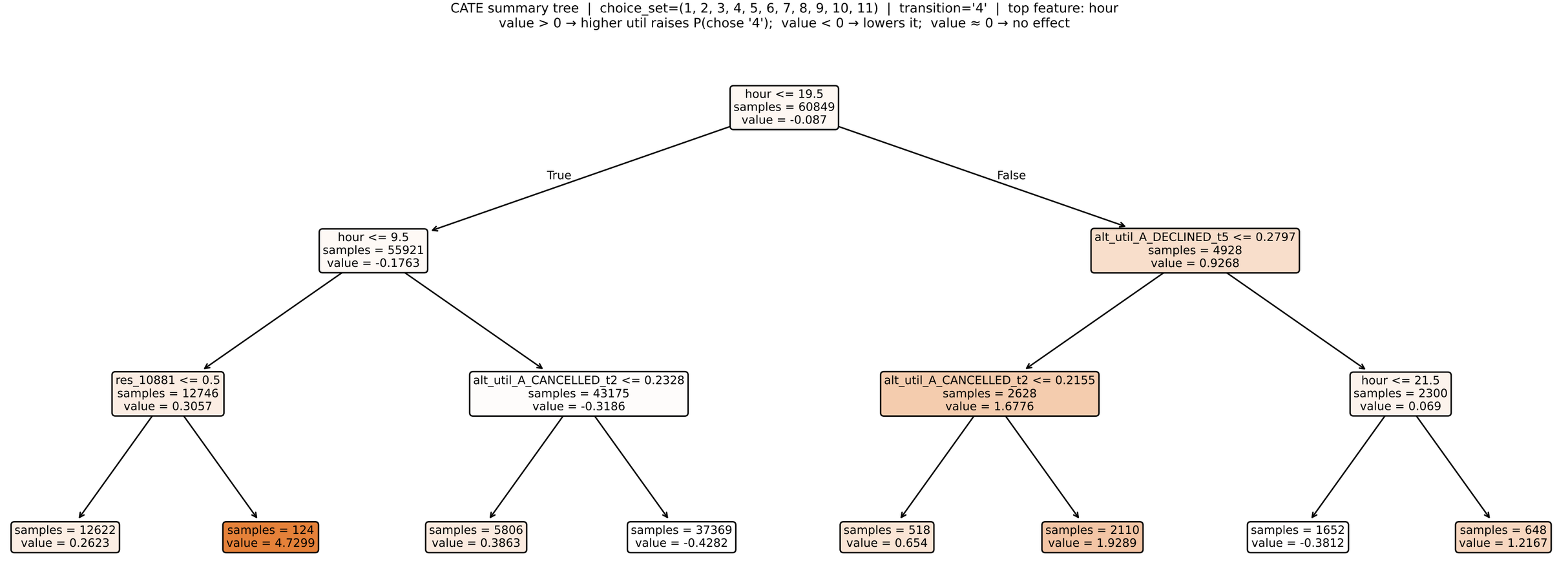

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='3' | top feature: hour
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_FINALIZED_t1 <= 0.5377
samples = 60849
value = -63.405
True
False
alt_util_A_REGISTERED_t9 <= 0.6111
samples = 60428
value = -67.2961
hour <= 9.5
samples = 421
value = 495.1016
hour <= 10.5
samples = 28920
value = -96.3577
hour <= 8.5
samples = 31508
value = -40.6215
hour <= 8.5
samples = 97
value = 1125.8349
hour <= 17.5
samples = 324
value = 306.2709
samples = 8617
value = -197.1419
samples = 20303
value = -53.5828
samples = 3957
value = 141.7073
samples = 27551
value = -66.8084
samples = 64
value = 1264.0646
samples = 33
value = 857.7531
samples = 227
value = 163.6787
samples = 97
value = 639.9661

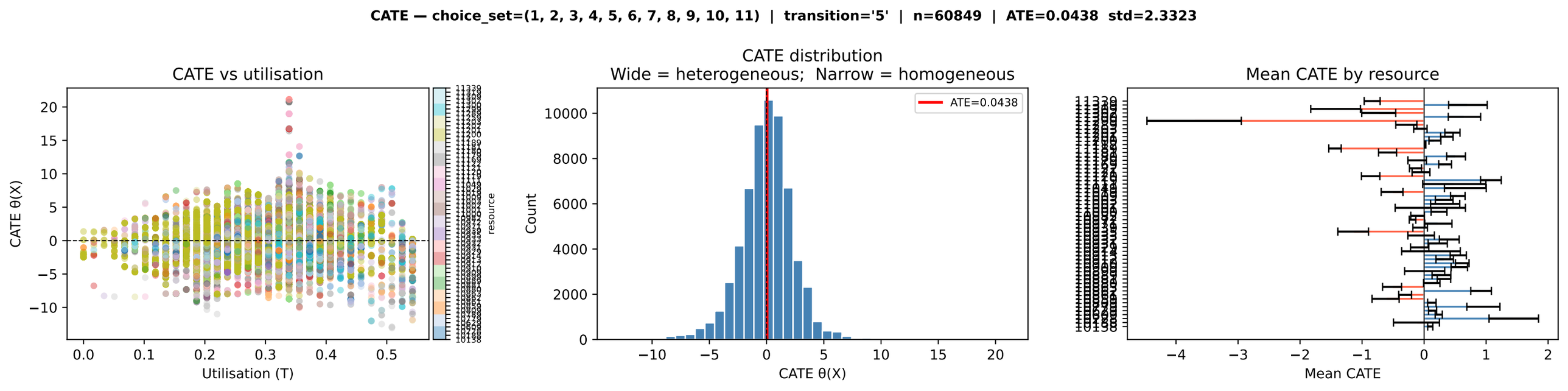
CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='4' | n=60849 | ATE=-0.0870 std=2.1630

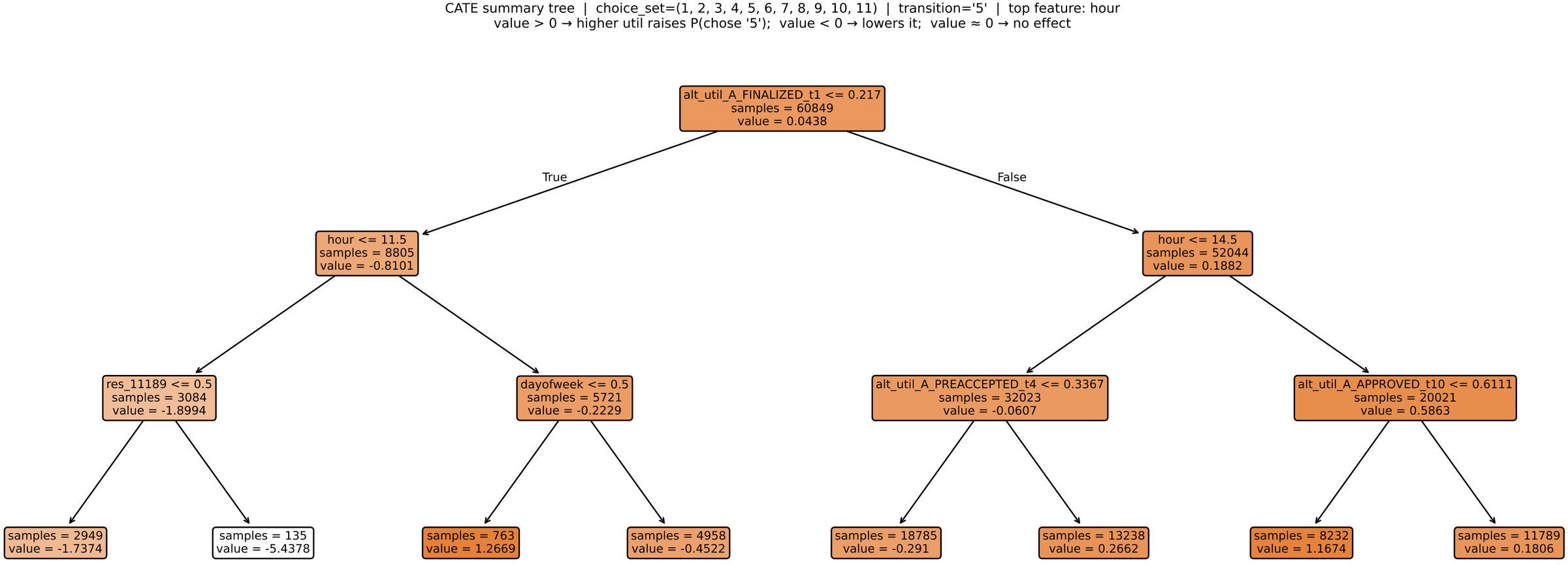


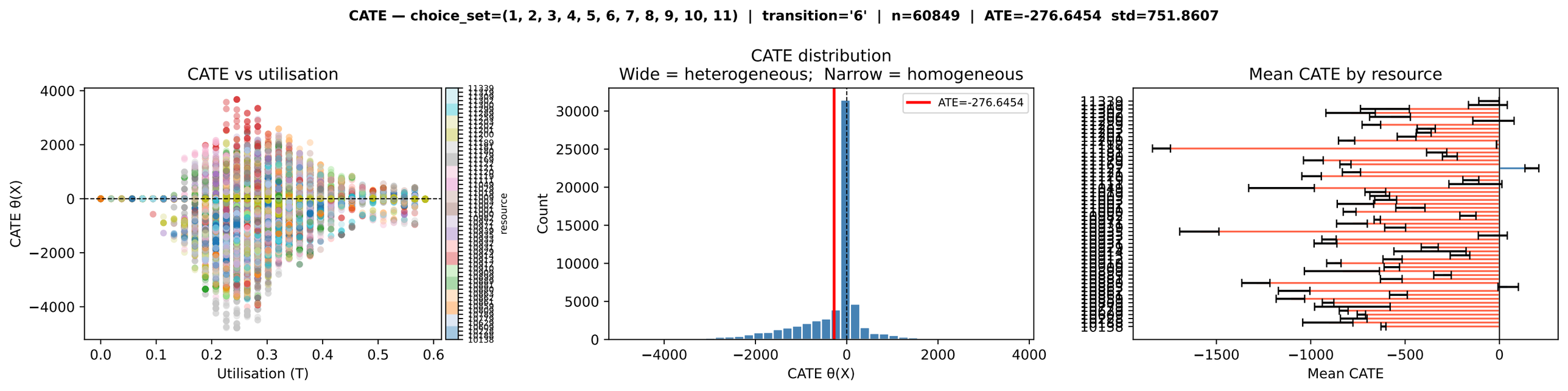


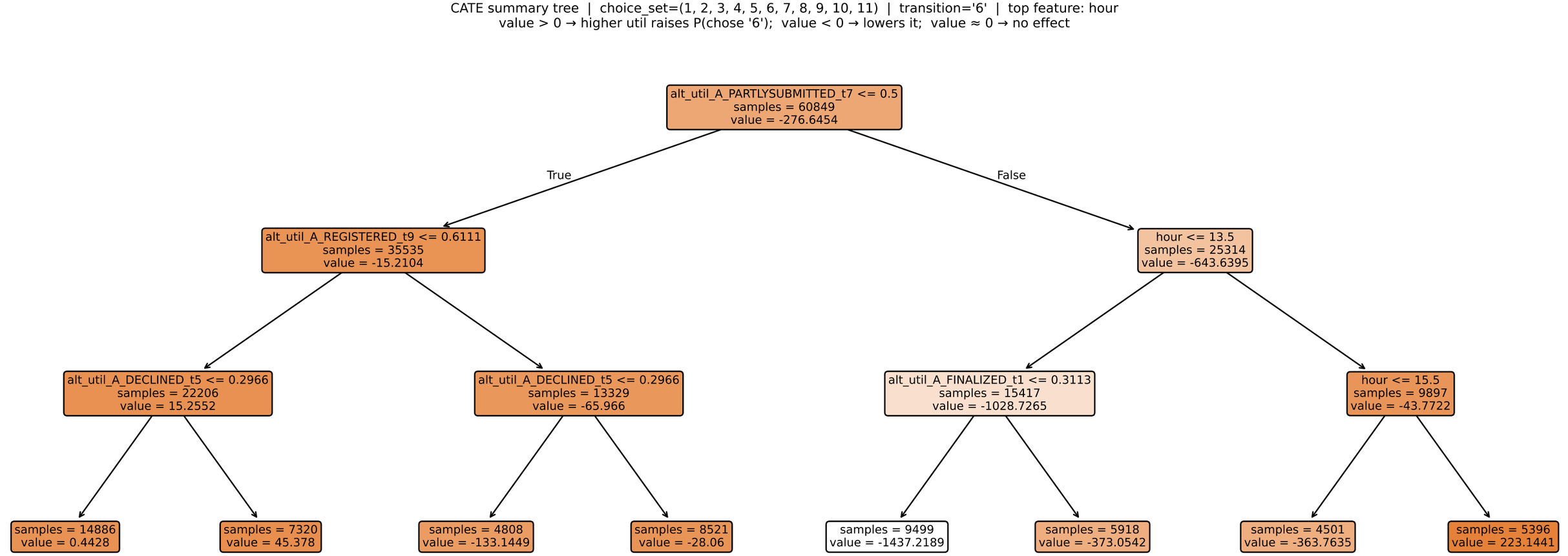



CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='4' | top feature: hour
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 19.5
samples = 60849
value = -0.087
True
False
hour <= 9.5
samples = 55921
value = -0.1763
alt_util_A_DECLINED_t5 <= 0.2797
samples = 4928
value = 0.9268
res_10881 <= 0.5
samples = 12746
value = 0.3057
alt_util_A_CANCELLED_t2 <= 0.2328
samples = 43175
value = -0.3186
alt_util_A_CANCELLED_t2 <= 0.2155
samples = 2628
value = 1.6776
hour <= 21.5
samples = 2300
value = 0.069
samples = 12622
value = 0.2623
samples = 124
value = 4.7299
samples = 5806
value = 0.3863
samples = 37369
value = -0.4282
samples = 518
value = 0.654
samples = 2110
value = 1.9289
samples = 1652
value = -0.3812
samples = 648
value = 1.2167

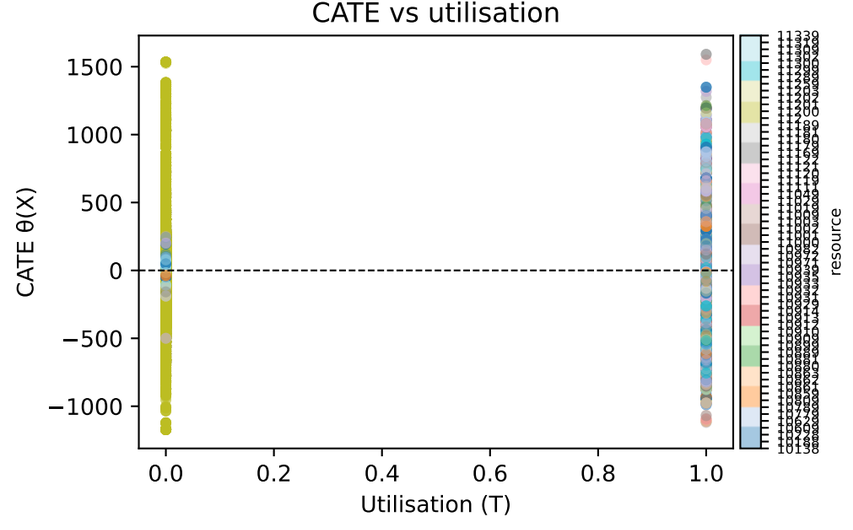

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='5' | n=60849 | ATE=0.0438 std=2.3323
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.0438
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

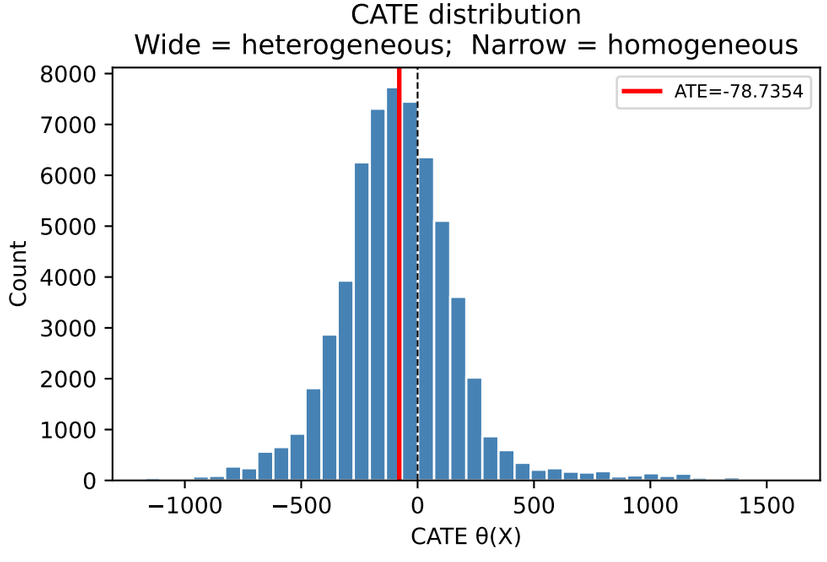

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='5' | top feature: hour
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_FINALIZED_t1 <= 0.217
samples = 60849
value = 0.0438
True
False
hour <= 11.5
samples = 8805
value = -0.8101
hour <= 14.5
samples = 52044
value = 0.1882
res_11189 <= 0.5
samples = 3084
value = -1.8994
dayofweek <= 0.5
samples = 5721
value = -0.2229
alt_util_A_PREACCEPTED_t4 <= 0.3367
samples = 32023
value = -0.0607
alt_util_A_APPROVED_t10 <= 0.6111
samples = 20021
value = 0.5863
samples = 2949
value = -1.7374
samples = 135
value = -5.4378
samples = 763
value = 1.2669
samples = 4958
value = -0.4522
samples = 18785
value = -0.291
samples = 13238
value = 0.2662
samples = 8232
value = 1.1674
samples = 11789
value = 0.1806

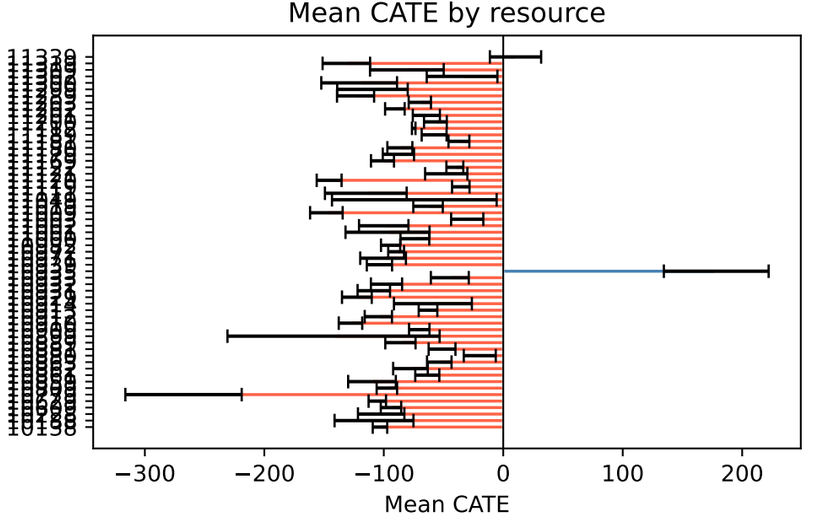

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='6' | n=60849 | ATE=-276.6454 std=751.8607
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-276.6454
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

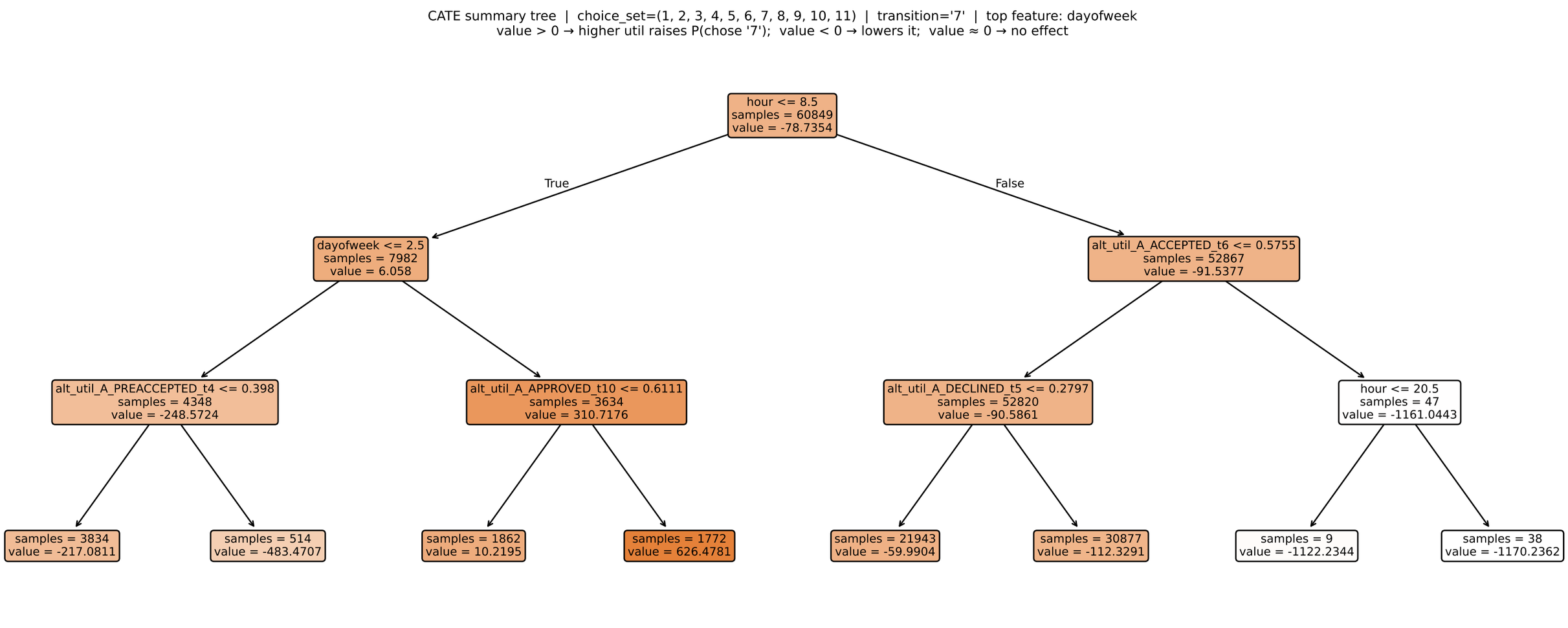

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='6' | top feature: hour
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_PARTLYSUBMITTED_t7 <= 0.5
samples = 60849
value = -276.6454
True
False
alt_util_A_REGISTERED_t9 <= 0.6111
samples = 35535
value = -15.2104
hour <= 13.5
samples = 25314
value = -643.6395
alt_util_A_DECLINED_t5 <= 0.2966
samples = 22206
value = 15.2552
alt_util_A_DECLINED_t5 <= 0.2966
samples = 13329
value = -65.966
alt_util_A_FINALIZED_t1 <= 0.3113
samples = 15417
value = -1028.7265
hour <= 15.5
samples = 9897
value = -43.7722
samples = 14886
value = 0.4428
samples = 7320
value = 45.378
samples = 4808
value = -133.1449
samples = 8521
value = -28.06
samples = 9499
value = -1437.2189
samples = 5918
value = -373.0542
samples = 4501
value = -363.7635
samples = 5396
value = 223.1441

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='7' | n=60849 | ATE=-78.7354 std=271.0919

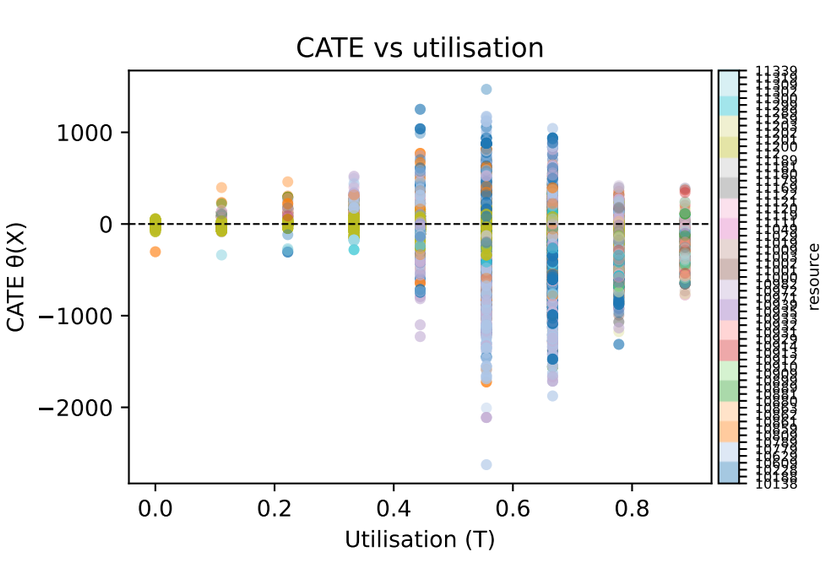


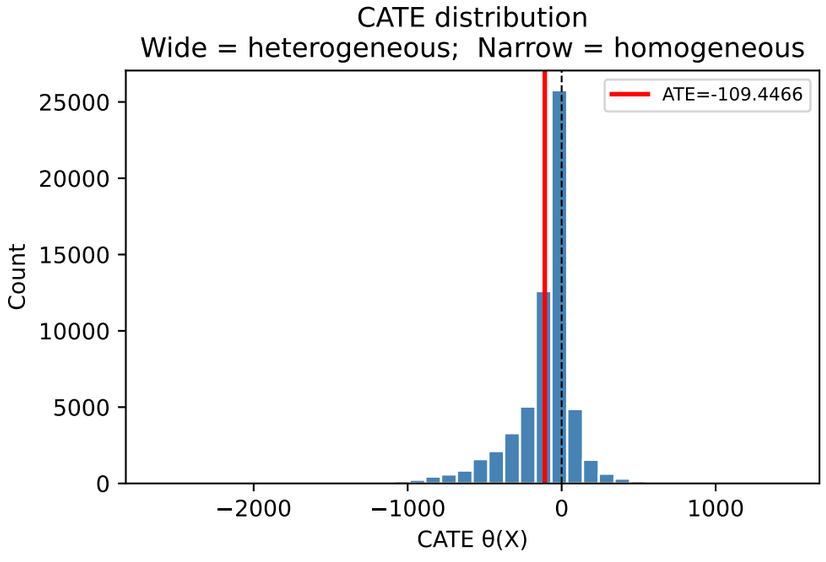


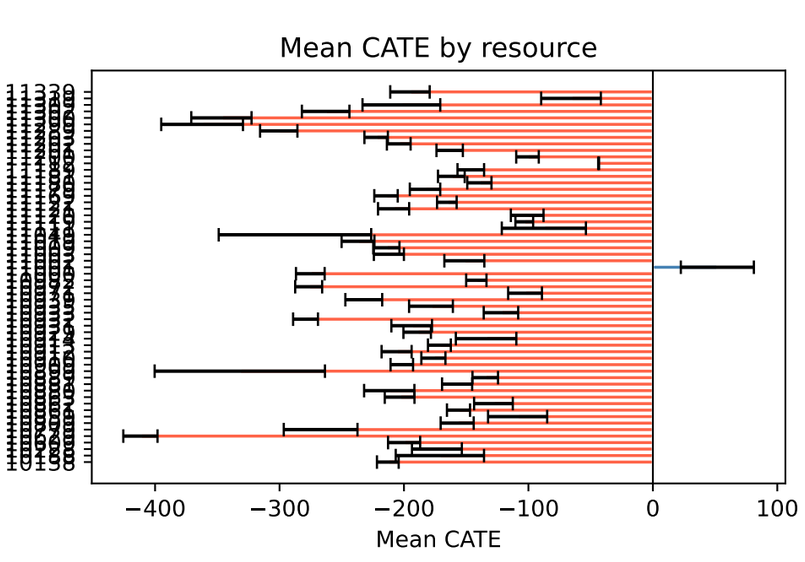


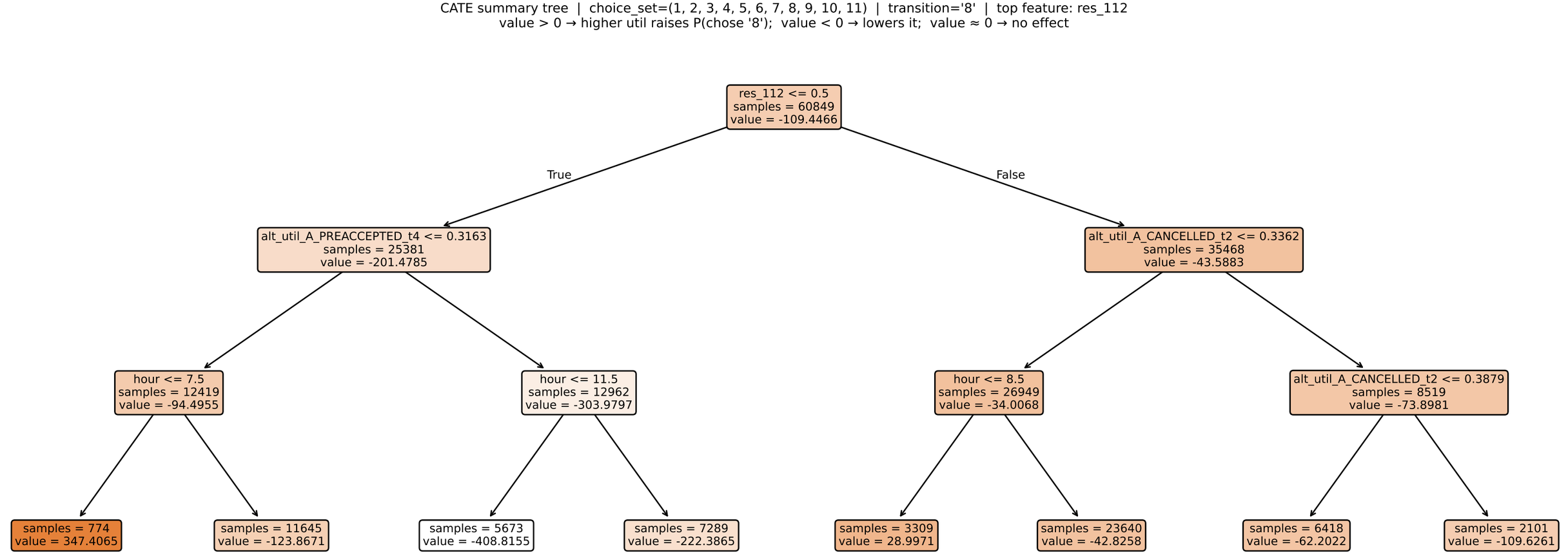

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='7' | top feature: dayofweek
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 8.5
samples = 60849
value = -78.7354
True
False
dayofweek <= 2.5
samples = 7982
value = 6.058
alt_util_A_ACCEPTED_t6 <= 0.5755
samples = 52867
value = -91.5377
alt_util_A_PREACCEPTED_t4 <= 0.398
samples = 4348
value = -248.5724
alt_util_A_APPROVED_t10 <= 0.6111
samples = 3634
value = 310.7176
alt_util_A_DECLINED_t5 <= 0.2797
samples = 52820
value = -90.5861
hour <= 20.5
samples = 47
value = -1161.0443
samples = 3834
value = -217.0811
samples = 514
value = -483.4707
samples = 1862
value = 10.2195
samples = 1772
value = 626.4781
samples = 21943
value = -59.9904
samples = 30877
value = -112.3291
samples = 9
value = -1122.2344
samples = 38
value = -1170.2362

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='8' | n=60849 | ATE=-109.4466 std=227.2965

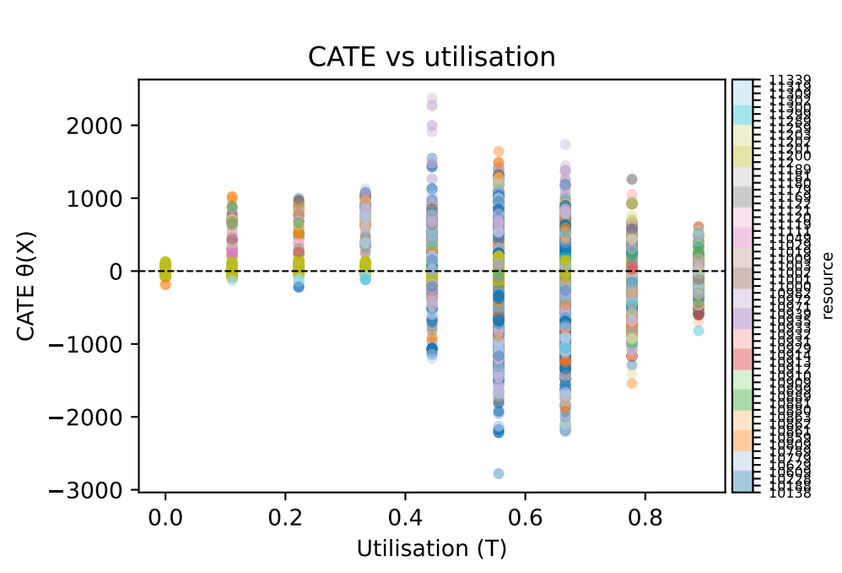


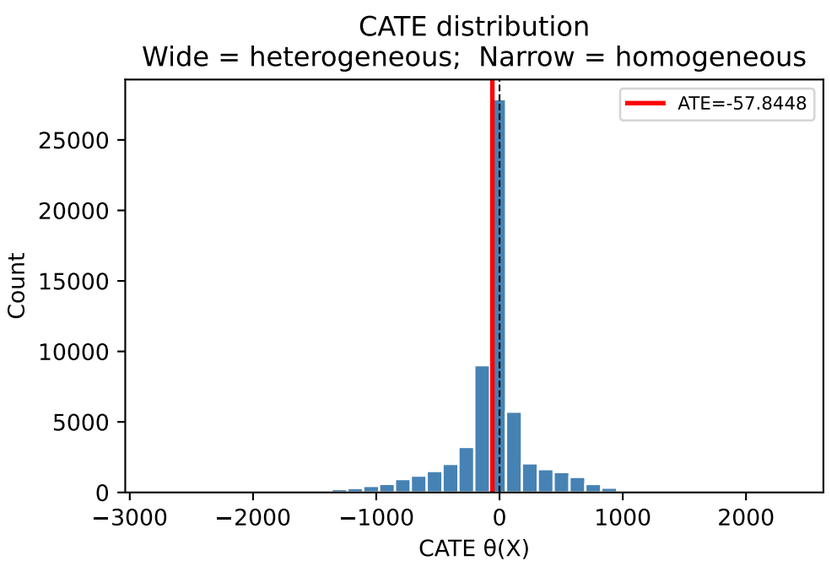


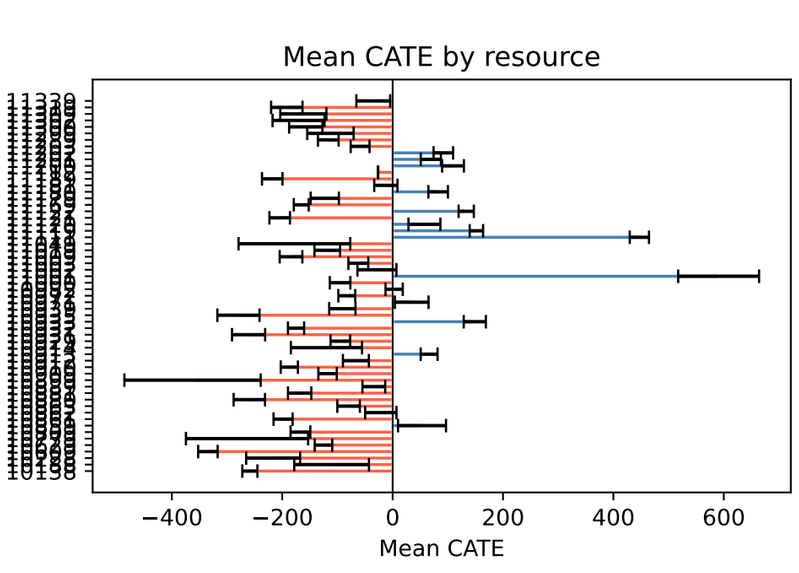


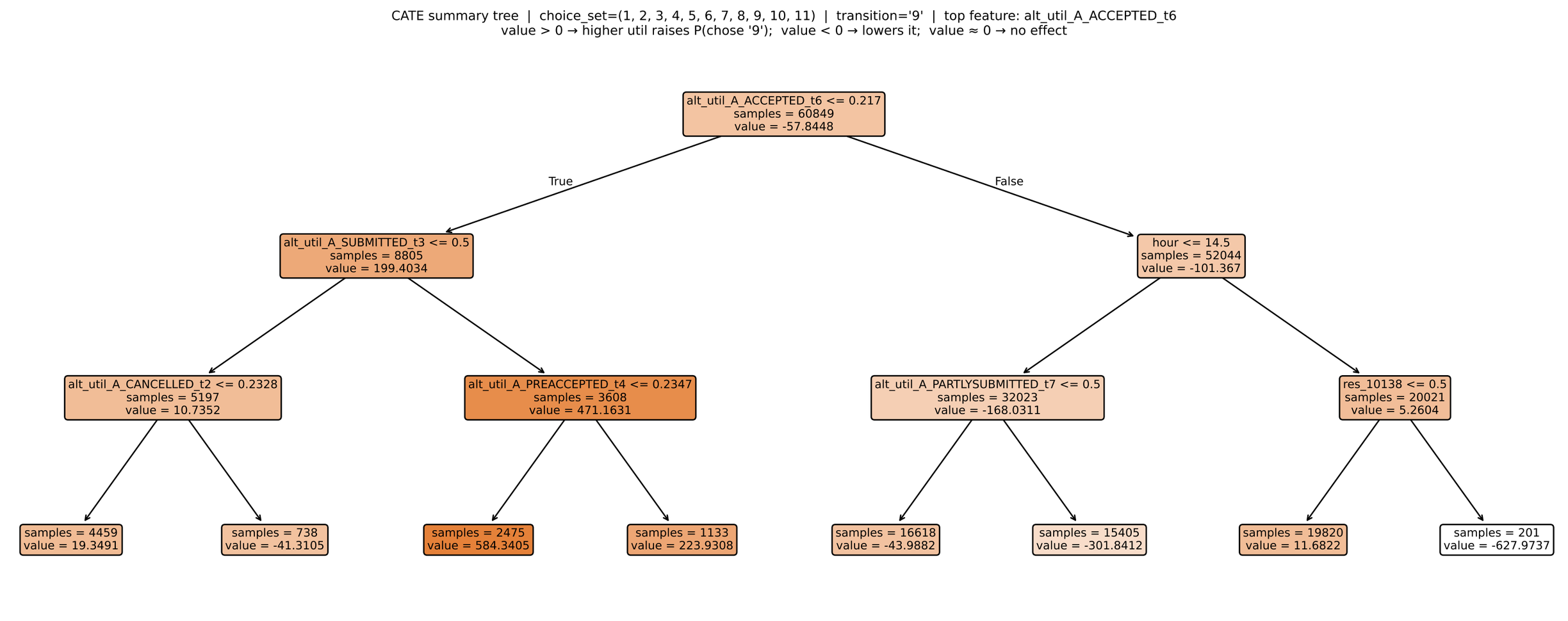

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='8' | top feature: res_112
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
res_112 <= 0.5
samples = 60849
value = -109.4466
True
False
alt_util_A_PREACCEPTED_t4 <= 0.3163
samples = 25381
value = -201.4785
alt_util_A_CANCELLED_t2 <= 0.3362
samples = 35468
value = -43.5883
hour <= 7.5
samples = 12419
value = -94.4955
hour <= 11.5
samples = 12962
value = -303.9797
hour <= 8.5
samples = 26949
value = -34.0068
alt_util_A_CANCELLED_t2 <= 0.3879
samples = 8519
value = -73.8981
samples = 774
value = 347.4065
samples = 11645
value = -123.8671
samples = 5673
value = -408.8155
samples = 7289
value = -222.3865
samples = 3309
value = 28.9971
samples = 23640
value = -42.8258
samples = 6418
value = -62.2022
samples = 2101
value = -109.6261

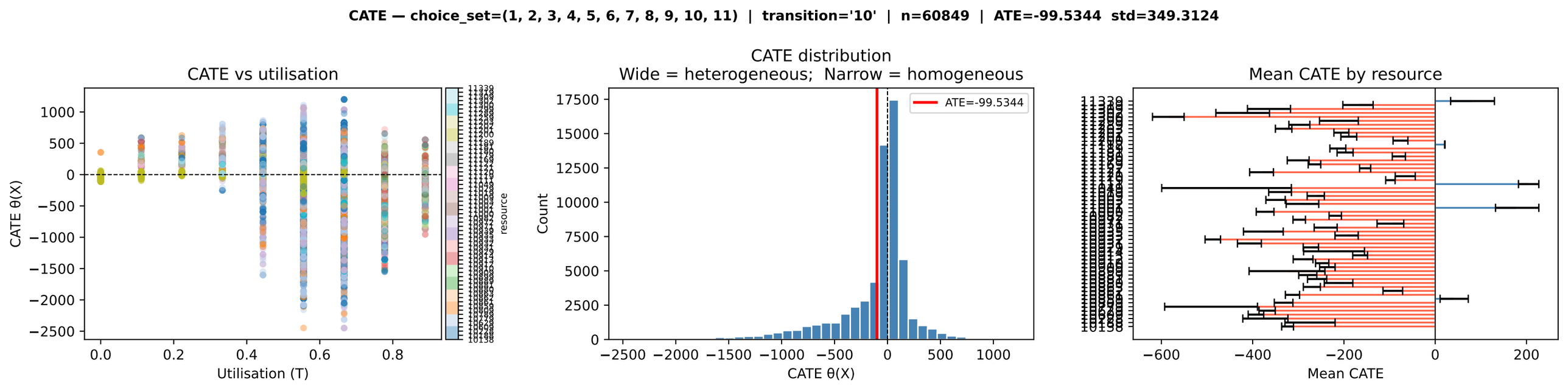
CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='9' | n=60849 | ATE=-57.8448 std=333.1160

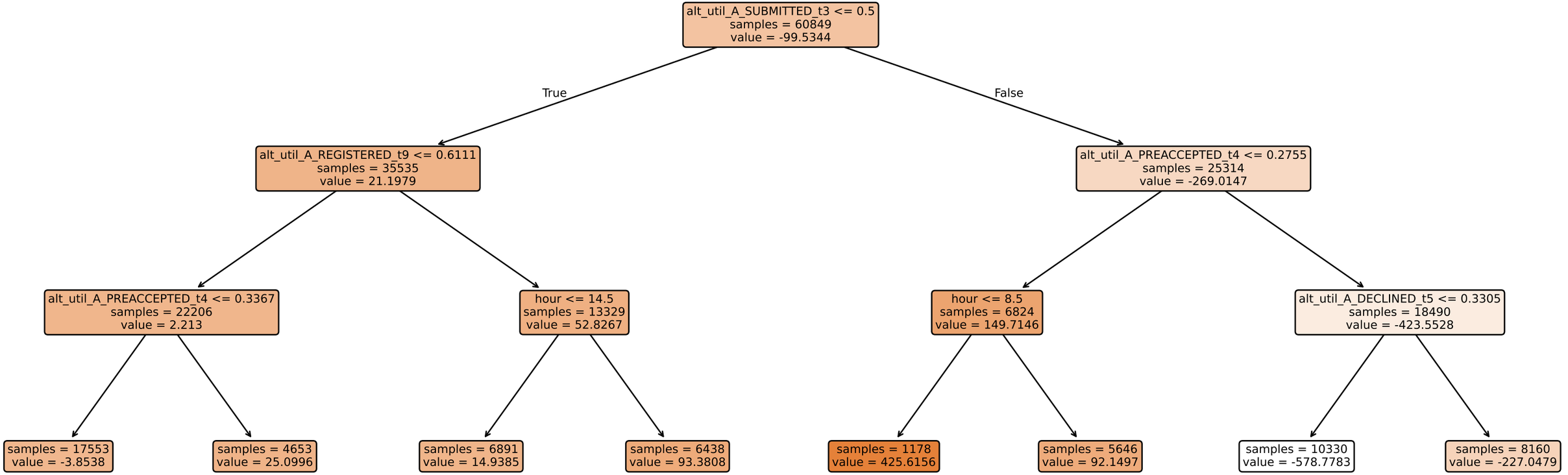







CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='9' | top feature: alt_util_A_ACCEPTED_t6
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_ACCEPTED_t6 <= 0.217
samples = 60849
value = -57.8448
True
False
alt_util_A_SUBMITTED_t3 <= 0.5
samples = 8805
value = 199.4034
hour <= 14.5
samples = 52044
value = -101.367
alt_util_A_CANCELLED_t2 <= 0.2328
samples = 5197
value = 10.7352
alt_util_A_PREACCEPTED_t4 <= 0.2347
samples = 3608
value = 471.1631
alt_util_A_PARTLYSUBMITTED_t7 <= 0.5
samples = 32023
value = -168.0311
res_10138 <= 0.5
samples = 20021
value = 5.2604
samples = 4459
value = 19.3491
samples = 738
value = -41.3105
samples = 2475
value = 584.3405
samples = 1133
value = 223.9308
samples = 16618
value = -43.9882
samples = 15405
value = -301.8412
samples = 19820
value = 11.6822
samples = 201
value = -627.9737


CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='10' | n=60849 | ATE=-99.5344 std=349.3124
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-99.5344
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) | transition='10' | top feature: alt_util_A_PREACCEPTED_t4
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_SUBMITTED_t3 <= 0.5
samples = 60849
value = -99.5344
True
False
alt_util_A_REGISTERED_t9 <= 0.6111
samples = 35535
value = 21.1979
alt_util_A_PREACCEPTED_t4 <= 0.2755
samples = 25314
value = -269.0147
alt_util_A_PREACCEPTED_t4 <= 0.3367
samples = 22206
value = 2.213
hour <= 14.5
samples = 13329
value = 52.8267
hour <= 8.5
samples = 6824
value = 149.7146
alt_util_A_DECLINED_t5 <= 0.3305
samples = 18490
value = -423.5528
samples = 17553
value = -3.8538
samples = 4653
value = 25.0996
samples = 6891
value = 14.9385
samples = 6438
value = 93.3808
samples = 1178
value = 425.6156
samples = 5646
value = 92.1497
samples = 10330
value = -578.7783
samples = 8160
value = -227.0479

Census Tract 9900.01, Washington County, Oregon
Subject | Washington County | Oregon | United States | Margin of error | ± | Percent | Percent margin of error | ± | Index (United States = 100) | Percent of total

Subject	Washington County	Oregon	United States	Margin of error	±	Percent	Percent margin of error	±	Index (United States = 100)	Percent of total
Total population	5,284	4,237,256	328,239,523	5,284	±	1.6%	0.0%	±	0.0%	0.0%
Male	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Female	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Median age (years)	39.2	39.1	38.2	39.2	±	1.6%	0.0%	±	0.0%	0.0%
Under 5 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
5 to 9 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
10 to 14 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
15 to 19 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
20 to 24 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
25 to 29 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
30 to 34 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
35 to 39 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
40 to 44 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
45 to 49 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
50 to 54 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
55 to 59 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
60 to 64 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
65 to 69 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
70 to 74 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
75 to 79 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
80 to 84 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
85 years and over	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Race										
White alone	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Black or African American alone	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
American Indian and Alaska Native alone	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Asian alone	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Native Hawaiian and Other Pacific Islander alone	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Some other race alone	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Two or more races	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Two races including Some other race	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Two races excluding Some other race, and three or more races	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Hispanic or Latino origin (of any race)	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
White alone, not Hispanic or Latino	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Population reporting ancestry	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Ancestry										
Arab	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Czech	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Danish	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Dutch	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
English	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
French (except Basque)	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
French Canadian	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
German	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Greek	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Hungarian	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Irish	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Italian	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Lithuanian	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Norwegian	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Polish	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Portuguese	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Russian	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Scotch-Irish	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Scottish	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Slovak	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Subsaharan African	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Swedish	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Swiss	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Ukrainian	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
United States or American	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Welsh	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
West Indian (except Hispanic groups)	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Other ancestries	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Ancestry unclassified	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Language spoken at home										
English only	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Language other than English	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Spanish	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Spanish: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
French (incl. Patois, Cajun)	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
French (incl. Patois, Cajun): Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
German or other West Germanic languages	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
German or other West Germanic languages: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Russian, Polish, or other Slavic languages	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Russian, Polish, or other Slavic languages: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Other Indo-European languages	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Other Indo-European languages: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Korean	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Korean: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Chinese (incl. Mandarin, Cantonese)	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Chinese (incl. Mandarin, Cantonese): Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Vietnamese	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Vietnamese: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Tagalog (incl. Filipino)	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Tagalog (incl. Filipino): Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Other Asian languages	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Other Asian languages: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Navajo	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Navajo: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Other Native North American languages	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Other Native North American languages: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Hungarian	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Hungarian: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Arabic	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Arabic: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Hebrew	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Hebrew: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
African languages	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
African languages: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Other and unspecified languages	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Other and unspecified languages: Speak English less than "very well"	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Total households	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Family households (families)	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
With own children under 18 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Married-couple family	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
With own children under 18 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Male householder, no spouse present	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
With own children under 18 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Female householder, no spouse present	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
With own children under 18 years	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Nonfamily households	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Householder living alone	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Male	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
65 years and over	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Female	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
65 years and over	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Households with one or more people under 18 years	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Households with one or more people 65 years and over	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Average household size	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Average family size	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Total housing units	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Occupied housing units	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Vacant housing units	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
For rent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Rented, not occupied	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
For sale only	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Sold, not occupied	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
For seasonal, recreational, or occasional use	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
For migrant workers	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Other vacant	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Homeowner vacancy rate	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Rental vacancy rate	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Housing tenure										
Occupied housing units	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Owner-occupied housing units	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Population in owner-occupied housing units	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Average household size of owner-occupied housing units	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Renter-occupied housing units	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Population in renter-occupied housing units	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Average household size of renter-occupied housing units	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Residence 1 year ago										
Population 1 year and over	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Same house	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Different house in the U.S.	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Same county	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Different county	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Same state	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Different state	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Abroad	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Value										
Owner-occupied housing units	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Less than $50,000	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$50,000 to $99,999	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$100,000 to $149,999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$150,000 to $199,999	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$200,000 to $299,999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$300,000 to $499,999	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$500,000 to $999,999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$1,000,000 or more	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Median (dollars)	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Mortgage status										
Owner-occupied housing units	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Housing units with a mortgage	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Housing units without a mortgage	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Selected monthly owner costs										
Housing units with a mortgage	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Less than $500	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$500 to $999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$1,000 to $1,499	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$1,500 to $1,999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$2,000 to $2,499	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$2,500 to $2,999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$3,000 or more	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Median (dollars)	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Housing units without a mortgage	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Less than $250	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$250 to $499	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$500 to $749	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$750 to $999	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$1,000 or more	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Median (dollars)	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Selected monthly owner costs as a percentage of household income in the past 12 months										
Housing units with a mortgage	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Less than 20.0 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
20.0 to 24.9 percent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
25.0 to 29.9 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
30.0 to 34.9 percent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
35.0 percent or more	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Not computed	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Housing units without a mortgage	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Less than 10.0 percent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
10.0 to 14.9 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
15.0 to 19.9 percent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
20.0 to 24.9 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
25.0 to 29.9 percent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
30.0 to 34.9 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
35.0 percent or more	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Not computed	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Gross rent										
Occupied units paying rent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Less than $500	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$500 to $999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$1,000 to $1,499	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$1,500 to $1,999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$2,000 to $2,499	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
$2,500 to $2,999	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
$3,000 or more	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Median (dollars)	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
No rent paid	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Gross rent as a percentage of household income in the past 12 months										
Occupied units paying rent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Less than 15.0 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
15.0 to 19.9 percent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
20.0 to 24.9 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
25.0 to 29.9 percent	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
30.0 to 34.9 percent	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
35.0 percent or more	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Not computed	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Total population	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
In group quarters	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Institutionalized population	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Male	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Female	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Noninstitutionalized population	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Male	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Female	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Total population	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
In group quarters	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Institutionalized population	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Male	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Female	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Noninstitutionalized population	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%	±	0.0%	0.0%
Male	2,669	2,137,110	167,217,764	2,669	±	1.6%	0.0%	±	0.0%	0.0%
Female	2,615	2,100,146	161,021,759	2,615	±	1.6%	0.0%

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	1	60849	-321.041141	649.428611	-3998.81675	-430.870639	-11.920326	0.094646	2661.579352	29.4	70.6	alt_util_A_SUBMITTED_t3
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	2	60849	0.091764	1.890859	-18.602587	-0.084818	0.001273	0.235369	20.378294	52.4	47.6	hour
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	3	60849	-63.40497	285.28637	-1250.359915	-239.082171	-71.351305	91.975304	1483.539691	38.6	61.4	hour
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	4	60849	-0.086973	2.163017	-15.533151	-1.35255	-0.082407	1.176397	9.968677	47.9	52.1	hour
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	5	60849	0.043774	2.332331	-13.083032	-1.238781	0.11947	1.369353	21.125054	52.3	47.7	hour
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	6	60849	-276.645357	751.860725	-4789.483792	-419.268115	-0.243605	0.503167	3675.431016	32.5	67.5	hour
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	7	60849	-78.73539	271.091886	-1172.474403	-230.318263	-88.325145	58.387014	1591.064322	34.1	65.9	dayofweek
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	8	60849	-109.446553	227.296481	-2625.385436	-160.635226	-55.332829	-5.892456	1469.271411	22.5	77.5	res_112
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	9	60849	-57.844761	333.116047	-2779.276633	-117.043306	-25.264884	34.251853	2370.286833	37.9	62.1	alt_util_A_ACCEPTED_t6
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	10	60849	-99.534407	349.312428	-2449.919854	-158.571927	4.427209	68.51818	1199.242862	52.2	47.8	alt_util_A_PREACCEPTED_t4

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md